Unbounded covering-packing ratios of three-dimensional lattices

Disproof of Conjecture 00000005609

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Abstract

For $t \geq 1$, the rectangular Euclidean lattice $\Lambda_t = t\mathbb{Z} \times \mathbb{Z} \times \mathbb{Z}$ has packing radius $1/2$ and covering radius $\sqrt{t^2+2}/2$. Its covering-packing gap is therefore $\sqrt{t^2+2}$, which is unbounded. This disproves the assertion of an upper endpoint attained by a cubic lattice. We do not assess the separate BCC minimum claim. Lean verifies actual Euclidean lattices and both radius extrema, including discreteness and full rank.

Definitions and the rectangular family

For a full-rank Euclidean lattice $\Lambda \subset \mathbb{R}^3$, its packing radius is

$$r_{\mathrm{pack}}(\Lambda) = \frac{1}{2} \min_{0 \neq v \in \Lambda} \|v\|_2.$$

Equivalently it is the largest radius for which the open balls at distinct lattice points are disjoint. Its covering radius is the least $r \geq 0$ such that the closed radius-r balls about its points cover all of $\mathbb{R}^3$. The source defines the gap as the ratio of covering radius to packing radius:

$$g(\Lambda) = r_{\mathrm{cover}}(\Lambda)/r_{\mathrm{pack}}(\Lambda).$$

For every real $t \geq 1$ take

$$\Lambda_t = \{(tm, n, p) : m, n, p \in \mathbb{Z}\}.$$

It is an additive subgroup generated by the three linearly independent vectors

$$(t, 0, 0), \qquad (0, 1, 0), \qquad (0, 0, 1).$$

Thus its real span is all of $\mathbb{R}^3$. Every nonzero lattice vector has a coordinate of absolute value at least one, hence has Euclidean norm at least one. Differences of distinct lattice points have the same property. The lattice is discrete, with distinct points at distance at least one. Moreover, $(0, 1, 0) \in \Lambda_t$ has norm one. Consequently

$$r_{\mathrm{pack}}(\Lambda_t) = \frac{1}{2}.$$

This proves both admissibility as a full-rank discrete lattice and the packing extremum directly.

The exact covering radius and unbounded gap

Given any $x = (x_0, x_1, x_2) \in \mathbb{R}^3$, choose integers m, n, p nearest to $x_0/t, x_1, x_2$ respectively. Then the three coordinate errors to (tm, n, p) have absolute values at most $t/2, 1/2, 1/2$. Therefore

$$\|x - (tm, n, p)\|_2^2 \le \frac{t^2 + 2}{4}, \qquad r_{\mathrm{cover}}(\Lambda_t) \le \frac{\sqrt{t^2 + 2}}{2}.$$

For the reverse inequality use the centre $c_t = (t/2, 1/2, 1/2)$. For every integer k we have $|1/2 - k| \ge 1/2$. Hence, for every $(tm, n, p) \in \Lambda_t$,

$$\begin{aligned} \|c_t - (tm, n, p)\|_2^2 &= t^2(1/2 - m)^2 + (1/2 - n)^2 + (1/2 - p)^2 \\ &\ge \frac{t^2 + 2}{4}. \end{aligned}$$

Any radius whose lattice balls cover c_t must be at least $\sqrt{t^2 + 2}/2$. The upper bound is therefore attained, giving

$$r_{\mathrm{cover}}(\Lambda_t) = \frac{\sqrt{t^2 + 2}}{2}, \qquad g(\Lambda_t) = \sqrt{t^2 + 2}.$$

Theorem 1. *The covering-packing gap over full-rank Euclidean lattices in dimension three is unbounded above.*

Proof. For any real B, set $t = \max(1, B + 1)$. Then $t \ge 1$ and $\sqrt{t^2 + 2} > t \ge B + 1 > B$. The displayed exact gap formula supplies an admissible lattice with gap larger than B. □

Thus the spectrum cannot have a finite upper endpoint attained by a cubic lattice, or by any lattice. For example the simple cubic lattice Λ_1 has gap $\sqrt{3}$, whereas Λ_4 has gap $\sqrt{18}$. Unboundedness alone is not asserted to rule out a closed half-infinite interval: it refutes the explicit attained upper-endpoint clause. The separate assertions concerning the BCC minimum and its value are not assessed. The source has no restriction on aspect ratio or well-roundedness. No such restriction is assumed here.

Formalization and reproducibility

Lean uses the actual Mathlib type `EuclideanSpace R (Fin 3)` and its Euclidean norm. The map `point t` sends integer triples to (tm, n, p); `latticeGroup` gives its actual additive subgroup. Theorems `point_injective`, `lattice_discrete` and `lattice_full_rank` prove its integer coordinate representation, actual discrete subtype topology and full real span for every $t \geq 1$.

The formal packing radius is the supremum of the set of nonnegative r satisfying $2r \leq \|v\|$ for every nonzero lattice vector. The theorem `packing_geometric` proves equivalence to the pairwise-distance condition for distinct lattice points. The formal covering radius is the infimum of the set of nonnegative r for which every point of the actual Euclidean space is within distance r of some lattice point. No explicit radius formula is inserted as a definition or assumed.

Theorems `packing_exact` and `covering_exact` identify these extrema. The covering proof uses integer rounding for the upper bound and the half-integer centre for a bound against *every* lattice point. Theorem `valid_lattice_unbounded` includes both the discrete-topology and full-rank certificates in each unboundedness witness; `no_upper_endpoint` directly negates a uniform finite upper bound. The cubic comparison is also checked.

Lean 4.19.0 and Mathlib revision `c44e0c8ee63ca166450922a373c7409c5d26b00b` are pinned. From the submitted `lean/` directory:

```
lake exe cache get
lake build
lake env lean -DwarningAsError=true Main.lean
```

The complete project is built in a fresh directory. The strict check treats warnings as errors; the axiom audit contains only standard foundational axioms. There are no placeholders, custom axioms, numerical approximations or external computation oracles. No auxiliary program is needed for this exact proof. Compile the document with `tectonic main.tex`.

The exact bilingual statement, commit-pinned provenance, build logs, hashes and PDF checks are included. An agent developed the proof and supplied a self-review; the parent separately reviewed the complete proof, semantics and final artifacts. No external independent review or novelty is claimed for this elementary lattice calculation.

References

[1] The Last Math Competition, Conjecture 00000005609. Exact bilingual source and commit-pinned provenance are included as `SOURCE.md` and its verification record.